ABESEC Ghaziabad
Department of Computer Science & Engineering

SYNOPSIS REPORT
(Session 2026-27)

Project Title: Lightweight Edge-AI Model for Plant Disease Classification				
Project Type (application, product, research, review etc.)		Research-oriented Application		
	Name	**Roll Number**	**Section**	**Signature**
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Signature		**Remarks:**		
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1. Problem Introduction

1.1 Motivation

Manual identification of plant diseases is labour-intensive, subjective, and often inaccurate, especially for small and marginal farmers who lack easy access to agricultural experts. Deep learning models such as CNNs can classify plant leaf diseases with high accuracy, but state-of-the-art models are typically large (25–100+ MB) and computationally expensive, making them unsuitable for deployment on low-cost edge devices such as smartphones and Raspberry Pi boards that are realistically available in rural settings. This project is motivated by the need for an accurate yet lightweight disease-detection model that can run efficiently on resource-constrained edge hardware, enabling affordable, offline, real-time crop-disease diagnosis for farmers.

1.2 Project Objective

The primary objective is to design, train, and optimize a lightweight deep learning model for plant leaf disease classification using model compression techniques — pruning, quantization, and knowledge distillation — so that inference can run efficiently on edge devices without significant loss in classification accuracy. The specific objectives are:

- Train a baseline CNN (MobileNetV2, transfer learning) on the PlantVillage dataset.
- Apply structured pruning and post-training quantization to reduce model size and latency.
- Apply knowledge distillation to transfer learning from the baseline (teacher) model to a compact student model.
- Benchmark accuracy, model size, and CPU inference latency of the compressed model against the baseline to simulate edge-device performance.

1.3 Scope of the Project

The scope of this project is limited to image-based classification of plant leaf diseases using the publicly available PlantVillage dataset covering multiple crop species and disease classes. The work covers model training, compression (pruning, quantization, knowledge distillation), and CPU-based benchmarking as a proxy for edge-device deployment, implemented entirely on Google Colab at zero financial cost. Deployment on physical edge hardware (e.g., Raspberry Pi, mobile app) and multi-crop field trials are treated as future extensions and are outside the current scope.

2. Related Previous Work

Prior research has extensively used CNN architectures such as VGGNet, ResNet, and Inception for plant disease classification on datasets like PlantVillage, reporting accuracies above 95%, but these models remain computationally heavy and memory-intensive. Separately, model-compression literature — including pruning (Han et al., 2016), quantization (Jacob et al., 2018), and knowledge distillation (Hinton et al., 2015) — has demonstrated that model size can be reduced by 4–10x with minimal accuracy loss on general image-classification tasks. However, limited published work combines these compression techniques specifically for agricultural disease-detection models intended for edge deployment, which is the gap this project aims to address.

3. Software and Hardware Requirements

Software: Python 3.x, TensorFlow/Keras (or PyTorch), TensorFlow Lite (for post-training quantization), Google Colab (free GPU/TPU runtime), NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV.

Hardware: No dedicated hardware investment is required. Model training uses Google Colab's free-tier GPU; a standard laptop/PC (4GB+ RAM) is sufficient for coding, data preprocessing, and result analysis. CPU-only inference is used to benchmark feasibility on edge-class hardware at zero cost.

4. Proposed Method

Step 1 – Data collection & preprocessing: acquire the PlantVillage dataset; perform resizing, normalization, and augmentation.

Step 2 – Baseline model: train MobileNetV2 using transfer learning as the teacher/baseline classifier.

Step 3 – Pruning: apply structured/unstructured weight pruning to remove redundant filters and weights.

Step 4 – Quantization: apply post-training INT8 quantization using TensorFlow Lite.

Step 5 – Knowledge distillation: train a smaller student network guided by the soft labels of the baseline teacher model.

Step 6 – Evaluation: compare baseline vs. compressed models on accuracy, F1-score, model size (MB), and CPU inference latency.

Step 7 – Documentation: prepare results, comparison plots, and a research paper draft for conference/journal submission.

5. Deliverables

- A trained baseline (MobileNetV2) plant-disease classification model.
- Pruned, quantized, and knowledge-distilled compressed model variants.
- A comparative performance report: accuracy vs. model size vs. inference latency.
- Fully reproducible source code and Google Colab notebooks (zero-cost pipeline).
- A research paper draft suitable for conference/journal submission.

6. Stakeholders

Farmers and agricultural workers who need low-cost, real-time crop-disease diagnosis; agricultural extension officers and NGOs supporting rural farming communities; developers building mobile/embedded agri-tech applications; and the academic community, who benefit from a reproducible edge-AI benchmark for plant disease detection.

7. Gantt Chart

Week	Task	Deliverable
1–2	Literature review, dataset acquisition & preprocessing (PlantVillage)	Cleaned dataset

3–4	Baseline model training (MobileNetV2, transfer learning)	Trained baseline model
5	Apply pruning to baseline model	Pruned model
6	Apply post-training quantization (TFLite)	Quantized model
7	Knowledge distillation (student model training)	Distilled compact model
8	Benchmarking: accuracy, size, CPU latency	Comparison report
9–10	Result analysis & research paper drafting	Paper draft & final synopsis/report

8. References

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[5] Jacob, B., et al., "Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference," CVPR, 2018.

[6] PlantVillage Dataset — publicly available plant leaf image dataset, Kaggle/GitHub repository.